



Toward an experimental test for the finite-time wave function collapse

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ABSTRACT

In spite of recent conceptual and experimental advances, the ultimate nature of the wave function in quantum theory remains nebulous. Is it a mathematical device that describes our knowledge on a physical system or is it a physical field in any tenable sense? In this work we report on significant experimental steps toward a test on the reality of the wave function, which does not address the duality “epistemic *versus* ontic” in the usual way. Instead of considering inequalities derived from the partial indistinguishability of non-orthogonal states, we focus on the possibility of discriminating instantaneous reduction from finite-time wave function collapse. The former can only be associated with a Bayesian update, while the latter would be compatible with realistic interpretations. We employ non-maximally entangled photons which must be finely synchronized before detection. The required partial entanglement of the pairs is characterized by the violation they produce in the Clauser–Horne–Shimony–Holt inequality and the synchronization is obtained via time-scan measurements. This realization allows us to better understand the necessary instrumental conditions to the execution of a final, conclusive test.

1. Introduction

The recent discussions on the ontological character of the wave function illustrate the difficulties to place the metaphysical question “Does the quantum state exist?” in physically reasonable terms, i.e., amenable to experimental tests.

Presently, the discussion is dominated by a dichotomy: either the quantum state ψ is epistemic or it is ontic. If the former is true, ψ represents the knowledge on some aspects of reality, while the latter implies that ψ directly represents the microscopic state of affairs. As one would expect, it is possible to defend each of these two interpretations, depending on the adopted premises. There is, however, a recent tendency in favor of the ontic view, supported by theoretical [1,2] and experimental [3–5] results. However, it must be clear that each approach relies on different assumptions. For instance, in [1], it is assumed that any system has a “real physical state” objective and independent of the observer. Therefore, the epistemic-versus-ontic issue still stands as an open problem. Whether or not an uncontroversial answer to this question provides a satisfactory response to the question posed in the first paragraph is yet another nontrivial matter.

In this work we report on advances toward tests on the reality of the wave function with a distinct, conceptually independent approach. Rather than investigate the referred dichotomy, what is usually done

via inequalities arising from the impossibility to fully distinguish non-orthogonal states, we intend to explore the intimate connection between the reality of the wave function and the nature of its collapse. We will address this problem in the spirit proposed in [6] as a proof of principle and, later, in more realistic terms in [7]. It goes as follows. If the wave function represents only information on some aspects of the underlying physical reality, then an instantaneous collapse is not problematic, since it would represent a Bayesian informational update. On the other hand, if ψ is a physical field in any reasonable sense, then, the collapse should not be instantaneous. Within the model proposed in [7] it is possible to experimentally distinguish this time scale from the time required for the macroscopic specification of the results of a measurement (when the pointer goes to a determined position). It turns out that this distinction is indirect, inferred from the statistical distributions of polarization (or spin) measurements in entangled pairs. Here we will be concerned with polarization-entangled photon pairs.

Let us briefly introduce some key features of the reduction models that we will address [7], and discuss why a non-instantaneous collapse would slightly modify the standard statistical predictions of quantum mechanics, under special conditions. In [7] the experimental fact that any actual measurement has a finite duration, Δt , is explicitly considered. In addition, it is assumed that a microscopic event, a

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“hit”, precedes the macroscopic outcome resulting from a measurement. More specifically, it is assumed that the collapse is a process initiated by a “random hit” whose occurrence time is taken as a stochastic variable obeying a probability distribution, which, in the present case, is identified with the temporal intensity profile of the photon wave packet. A third assumption is that the quantum state takes a short time ($\delta t \ll \Delta t$), starting from the hit, to complete its reduction. We do not make any specific statements about this nonunitary time evolution. Now, if we consider, e.g., the state of polarization of two photons, which are sent to different detectors, and the two polarizations are measured during the same time window in the laboratory frame, then there is a small, but finite probability that the two hits (one for each detector) occur within a time interval shorter than δt . In these rare occasions, the second hit may change the outcome toward which the initial collapse would lead to. This would slightly change the frequencies of the measured polarizations, thus being experimentally accessible. For details on the model we refer to [7].

Apart from the fact that entanglement is required, there are not much similarities with Bell tests. In the first place, maximally entangled states are not useful for the present investigation, and it is necessary to produce unbalanced entangled states in a controlled way. In the polarization basis, we should have then states like $|\psi\rangle = \alpha|HV\rangle + \beta|VH\rangle$, with $\alpha \neq \beta$ and H (V) standing for horizontal (vertical) polarization. This has been accomplished with the use of twin photons, generated by parametric down conversion, passing through a Sagnac interferometer [8]. Secondly, the two parties have to measure the *same* observable, say σ_z , and the two measurements have to be finely synchronized in the laboratory reference frame. This synchronization requirement, absent in Bell tests, poses technical difficulties which are addressed here.

Finally, it is important to note that, what is proposed in [7] is not a detailed model for the wave-function collapse, but rather, a way to check the time scale of the reduction assuming that some stochastic mechanism starts it, without making detailed assumptions on the primary mechanisms and dynamics. Several other models are possible and have similarities with the assumptions in [7]. One well known model is the GRW spontaneous collapse theory [9] (for a more recent account including experimental investigations see [10]). In reference [11,12], the collapse is considered as a real physical process, however, without intrinsic stochasticity, with the probabilistic character of Born’s rule coming from unknown initial conditions. We refer to experiments investigating the state vector collapse as described in [7] as tests for Finite-Time Wave Function Collapse (FTWFC).

In the following we report a significant step in the implementation of the experimental test for FTWFC, as we perform the required synchronization procedure between two detectors using the necessary asymmetric states under our current best experimental conditions. In this way, we are now able to better estimate the conditions that need to be met for the final test of the proposal, and to conceive a detailed experiment for its implementation. Of course, the experimental setup and procedure that arise from this analysis is considerably more complex and demanding than the picture originally proposed in [7]. For example, Ref. [7] provides an analysis for a variable delay between two detection events, but assumed perfect synchronization of other auxiliary detections. In practice, all detections have to be synchronized and three delay lines will be required for the final test of the FTWFC, spanning a much larger set of time delays. As no specific experimental setup is assumed in Ref. [7], another difference is the absence of a discussion concerning the accumulation of systematic and random errors when compared to the expected statistical errors coming from projective measurements. In order to observe the statistical differences in detection probabilities predicted for FTWFC, the statistical errors have to prevail over all other sources of errors in the system, leading to strong constraints to be met by the experimental apparatus. In the end, we conclude that the final experimental test of FTWFC will be challenging, requiring state of the art performance of all elements in the setup, but it is still feasible with currently available technology.

In the next section we give a detailed account of the experimental setup employed in the present analysis. In Section 3 we characterize the two-photon polarization states in our experiment through their Bell non-locality, via standard Clauser–Horne–Shimony–Holt tests [13]. Section 4 presents our main experimental results of time-scan measurements. In Section 5 we discuss the results of Section 4 in light of the requirements to test the proposal of Ref. [7]. We also introduce in this section the new, more detailed proposal for the test of FTWFC. Our conclusions are presented in Section 6.

2. Experimental setup

The experimental apparatus used in the present investigation of the FTWFC is introduced schematically in Fig. 1. A mode-locked titanium-sapphire (Ti:S) laser with pulse duration $\tau_p \approx 100$ fs, repetition frequency $f_{rep} \approx 82$ MHz, and centered at $\lambda_c \approx 800$ nm is used to generate correlated photons via Parametric Down Conversion (PDC). The laser is frequency doubled by Type-I second-harmonic generation in a 500 μm barium borate (BBO) crystal. The beam at $\lambda_{SH} \approx 400$ nm is used as a pump for the entangled photons source. A dichroic mirror is used to separate λ_{SH} from the fundamental Ti:S laser beam at λ_p . A half-wave plate (HWP) and a quarter-wave plate (QWP) are used to control the polarization of the pump in order to manipulate the quantum state of the entangled photons.

A Sagnac interferometer with a periodically poled KTP crystal (PPKTP) is used as a source of entangled photons [14]. The PPKTP is grown for collinear, degenerate type II spontaneous parametric down conversion (SPDC) [15,16]. Such crystal has proven itself to be a highly efficient source for SPDC with no transverse walk-off between horizontal and vertical polarizations, due to the phenomenon of quasi-phase matching [17]. The pump beam is weakly focused to a beam waist of 92(2) μm at the crystal center, and has power of 22 mW before entering the Sagnac interferometer.

Inside the Sagnac, two optical elements are used for polarization control: a dual-wavelength (800 nm/400 nm) HWP (Casix) and a broadband polarizing beam splitter (PBS). The PBS is used to couple light in and out of the interferometer. The dual HWP has two functions. The first is to rotate the pump polarization in order to satisfy phase matching conditions in the crystal for both directions, clockwise and counterclockwise, see the Sagnac insert in Fig. 1. The second is to rotate the counterclockwise down-converted photon polarizations and in this manner ensure that the clockwise and counterclockwise idler photons exit the interferometer through the same output port (1 or 2), and the signal photons exit through the other one [8].

Once exiting the source, the entangled photons are sent through a polarization analysis module, composed of a HWP and a PBS. We detect both the reflected and transmitted beams from the PBS separately. Interference filters, centered at 800 nm and with FWHM of 10 nm (40 nm) are placed at the beams transmitted (reflected) by the two PBS, ensuring symmetry for the polarization analysis on output ports 1 and 2 of the Sagnac. For further spectral filtering, we use dichroic mirrors and long-pass color glasses.

Telescopes are placed in all detection channels to improve the coupling of the entangled photons to single-mode fibers. The fibers are connected to single-photon Si detectors (Perkin-Elmer SPCM). The photons from the Sagnac output ports 1 and 2, which are transmitted through the PBS analyzers, are coupled to 30 cm long fibers, while the reflections are coupled into 200 cm long fibers. A micrometric delay line is placed in the path of the Sagnac output port 1 transmission in order to perform the time-resolved measurements necessary to investigate the effects of FTWFC. As discussed in detail in Section 4, the size of the small fibers is an important ingredient to determine how much the delay line needs to be scanned. Note that it is not our goal here to synchronize the four detection outputs, as would be required to test the proposal of Ref. [7]. For that, it would be necessary to introduce delay lines and small fibers in the two reflection outputs of the PBS analyzers.

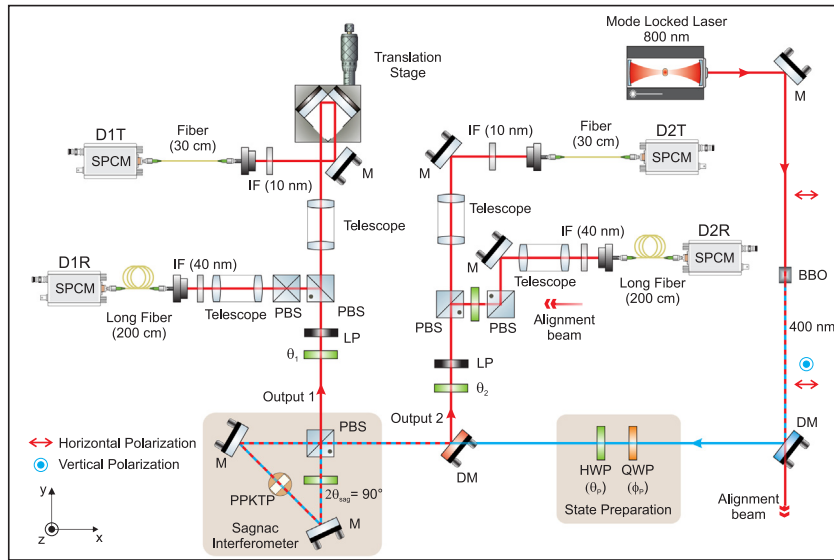


Fig. 1. (colors online) Experimental setup for synchronization of two detections on investigations of the finite-time wave function collapse. A Ti:sapphire laser beam is frequency doubled by a BBO crystal and employed as the pump beam for a source of entangled photons that uses a Sagnac interferometer. The Sagnac output ports are divided in two PBS, and in one arm is placed a delay line. SPCM, single photon counting module; LP, long pass filters; IF, interference filters; M, mirrors; DM, dichroic mirrors.

The electronic TTL signals from all detectors are sent to a NIN logic converter and processed by a coincidence count driver (FastCom TecP7888) with 1 ns resolution. In our scheme the pulse that triggers the coincidence detection is related to the D1T or D1R detectors (electronically delayed with respect to each other), and the two stop pulses are related to the D2T and D2R detectors. In this way, we can measure any coincidences between the four detectors. The efficiency of each detector is $\eta_{1T} = 4.3\%$, $\eta_{1R} = 2.5\%$, $\eta_{2T} = 3.5\%$, and $\eta_{2R} = 2.2\%$, as measured by the ratio between coincidences and single counts. They are defined then as conditional probabilities to detect a second event once a first detection occurs. We attribute the low detection efficiency mainly to the presence of a large number of fluorescence photons reaching the detection apparatus due to the interaction of the UV pump beam with optical elements, which results in an increase in the rate of single counts. Such fluorescence photons lead to false heralding events that decrease the overall conditional probability to detect a second photon. Time filtering could reduce such unwanted single counts due to fluorescence. Unfortunately, our acquisition time filtering only is available in the coincidence detections. For the purposes of the present work, only the coincidence counts are related to the photon-pair generation, which allowed us to proceed. The background coincidence counts due to fluorescence were measured by increasing the time window to 20 ns far from the SPDC arrival time. We register roughly 5 coincidence counts per second, whilst for the correlated photons we have 160 coincidences per second in a 4 ns time window.

3. Quantum states

The quantum state associated with the polarization degree of freedom when the down-converted photons exit the Sagnac can be written as [8]:

$$|\psi_{spdc}\rangle = \alpha|H_i, V_s\rangle + \beta e^{i(\phi_- + \Delta)}|V_i, H_s\rangle, \quad (1)$$

where i and s correspond to the idler and signal photons, respectively, with horizontal (H) or vertical (V) polarization, while α and β are the absolute values of the amplitudes

$$\alpha = \sqrt{\frac{\sin^2(2\theta_p) + \sin^2[2(\theta_p - \phi_p)]}{2}}, \quad (2)$$

$$\beta = -\sqrt{\frac{\cos^2(2\theta_p) + \cos^2[2(\theta_p - \phi_p)]}{2}}, \quad (3)$$

where θ_p and ϕ_p are the fast axis angles of the HWP and QWP, respectively, used in the pump-beam polarization control. In this derivation, we assume that the pump beam has polarization $|V\rangle$ before it passes through the HWP and QWP. The contribution ϕ_- to the relative phase between the two states in Eq. (1) is given by

$$\tan \phi_- = \frac{2 \sin(2\phi_p)}{\sin(4\theta_p) + \sin[4(\theta_p - \phi_p)]}. \quad (4)$$

Moreover, the constant Δ takes into account accumulated phase differences in the propagation of pump, signal and idler in the two opposite directions inside the Sagnac [8]. Experimentally, it is straightforward to find the polarization angles of the pump that leads to a maximally entangled state with $\phi_- + \Delta = 0$ [8]. We can define $\phi_p = 0^\circ$ for this position of the pump QWP [8]. In this case, Eq. (1) assumes the simpler form

$$|\psi_{spdc}\rangle = \alpha|H_i, V_s\rangle + \sqrt{1 - \alpha^2}|V_i, H_s\rangle, \quad (5)$$

with $\alpha = |\sin(2\theta_p)|$. A maximally entangled state is then obtained for $\alpha_M = 1/\sqrt{2}$, corresponding to $2\theta_p = 45^\circ$. Refs. [6,7] suggest the value $\alpha_C = 1/2$, corresponding to $2\theta_p = 30^\circ$, to maximize deviations corresponding to the FTWFC effect.

Eqs. (1)–(4) shows that by adequate manipulation of the optical waveplates in the pump beam it is possible to obtain the photonic state, Eq. (5), necessary to investigate the FTWFC dynamics proposed by Moreno and Parisio [7]. Such state should be a bipartite state with quantum correlations, but not maximally entangled [7]. From now on, we call this an unbalanced state. To proceed in this program our first set of measurements is designed to characterize the states produced by the source of entangled photons, which is accomplished by correlated polarization measurements in a set of distinct basis [18].

Figs. 2 and 3 show the coincidence counts acquired as we vary the HWP angle θ_2 in the output port 2 of the Sagnac interferometer. C_{xy} stands for the number of coincidences between the detectors x and y at the first and second output ports of the Sagnac, respectively. In both Figs. 2 and 3 we set $\phi_p = 0^\circ$ and squares are associated with α_M , while diamonds correspond to α_C . Fig. 2 presents the coincidence counts for

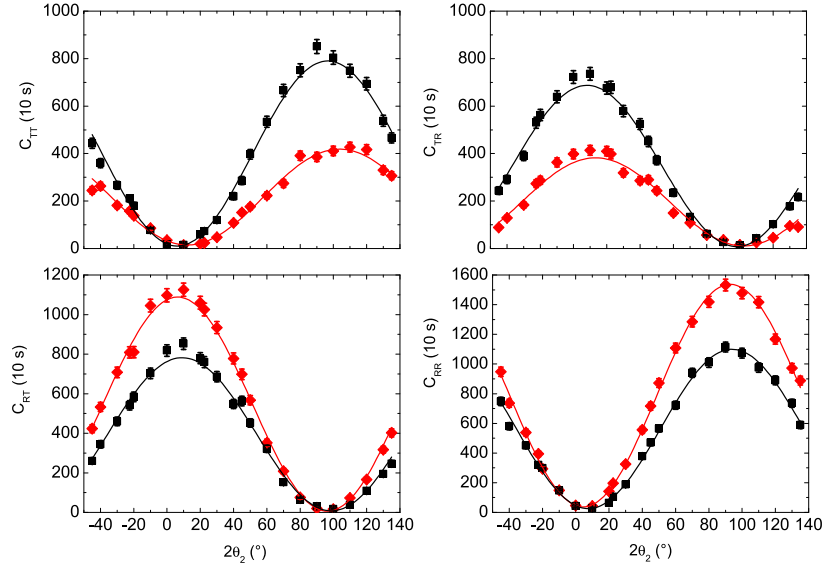


Fig. 2. (colors online) Coincidence counts as function of polarizing angle ($2\theta_2$) measured in the basis H/V ($\theta_1 = 0^\circ$) for α_M (squares) and α_C (diamonds). The error bars are given by the square roots of the count numbers. The solid line is a sinusoidal function that best represents the data. The symbol C_{xy} designates the number of coincidences between the detector x , in the first output port of the Sagnac, and the detector y , on the second output.

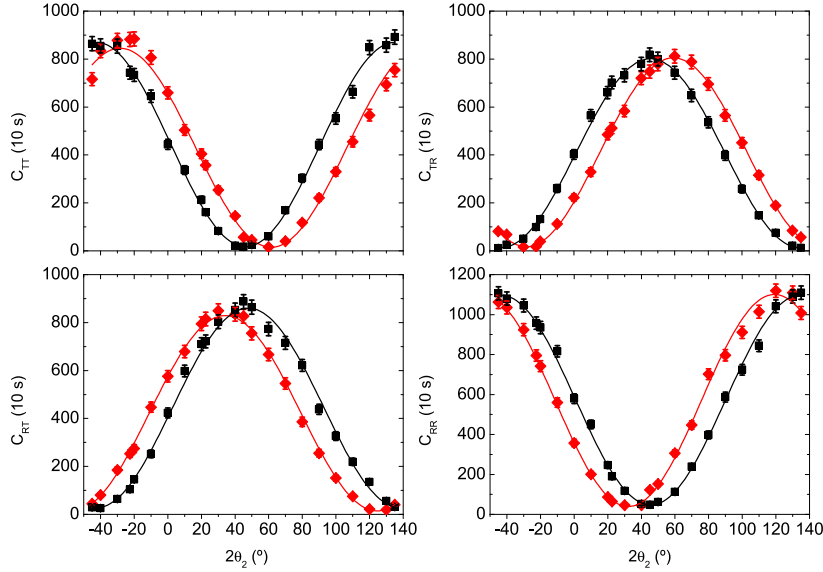


Fig. 3. (colors online) Coincidence counts as function of polarizing angle ($2\theta_2$) measured in the basis D/AD ($2\theta_1 = 45^\circ$) for α_M (squares) and α_C (diamonds). The error bars are given by the square roots of the count numbers. The solid line is a sinusoidal function that best represents the data. The symbol C_{xy} stands for the same quantity as in Fig. 2.

the H/V basis, while Fig. 3 presents the coincidence counts for the D/AD basis.

The solid lines represent the best sinusoidal fit to the experimental data, which is used to obtain the visibilities $V = (C_{\max} - C_{\min}) / (C_{\max} + C_{\min})$, with $C_{\max}$ and $C_{\min}$ the maximum and minimum values of coincidences, respectively. Table 1 shows the visibilities for the 16 experimental curves of Figs. 2 and 3.

One way to characterize entanglement in two-level bipartite systems is using the Clauser–Horne–Shimony–Holt (CHSH) inequality [13] stated in terms of the parameter

$$S = |E(\theta_1, \theta_2) + E(\theta_1, \theta'_2) + E(\theta'_1, \theta_2) - E(\theta'_1, \theta'_2)|, \quad (6)$$

where θ_1 and θ_2 are the angles of the HWP analyzers in Fig. 1 and $E(\theta_1, \theta_2)$ is defined as

$$E(\theta_1, \theta_2) = \frac{C_{TT} - C_{TR} - C_{RT} + C_{RR}}{C_{TT} + C_{TR} + C_{RT} + C_{RR}}. \quad (7)$$

C_{xy} designates the number of coincidences between the detector x , on the Sagnac first output port, and the detector y , on the second output port, with $x, y = T, R$.

States with quantum correlations present $2 < S < 2\sqrt{2} \approx 2.83$, and the maximal violation is achieved by a Bell state. For our case, the angle values that maximize the CHSH inequality violation are $\theta_1 = 0^\circ$, $2\theta'_1 = 45^\circ$, $2\theta_2 = 22.5^\circ$ and $2\theta'_2 = -22.5^\circ$. We use these values to measure S for the particular states defined by α_M and α_C . We obtain $S = 2.51 \pm 0.06$ for α_M and $S = 2.31 \pm 0.06$ for α_C , verifying that both are entangled states. More importantly, we are able to produce the entangled state α_C considered by Moreno and Parisio [7], necessary

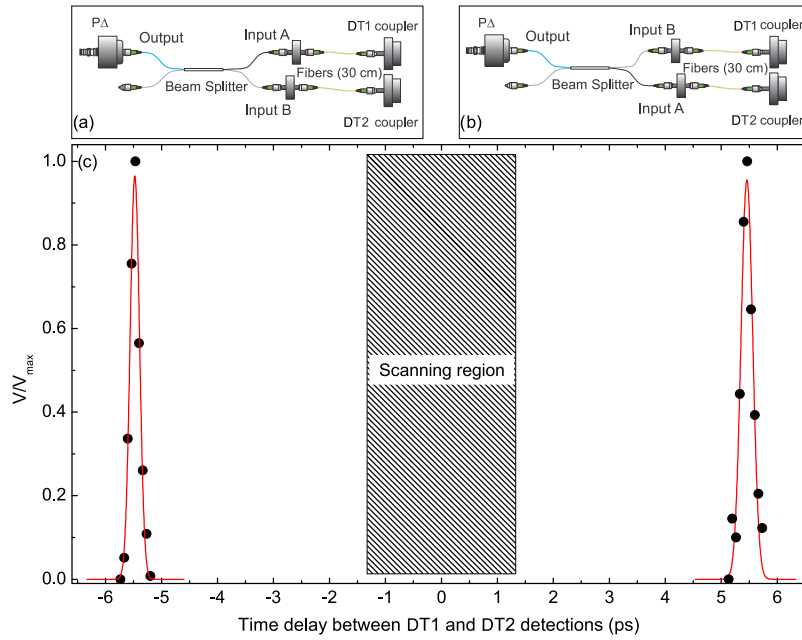


Fig. 4. (colors online) Panels (a) and (b) presents the experimental schemes measuring interference when the delay line marks -5.47 ps and $+5.47$ ps, respectively. Inputs A and B of the fiber beam splitter are coupled to the outputs of the two small, 30-cm fibers of Fig. 1. PD stands for Photo-Detector. (c) Interference measurement as function of the delay line position. The left peak is related to the experiment described in (a), while the right peak to the experiment in (b). With this, we define the scanning region from -1.3 to $+1.3$ ps.

Table 1

Visibility of the set of measures with interference in the base H/V ($\theta_1 = 0^\circ$) and D/AD ($2\theta_1 = 45^\circ$). The symbol $V_{x,y}$ represents the visibility between the detector x , in the first output port of the Sagnac, and the detector y , in the second output.

$2\theta_1$ ($^\circ$)	State	V_{TT} (%)	V_{TR} (%)	V_{RT} (%)	V_{RR} (%)
0	α_C	93.7(6.3)	93.0(6.4)	97.2(4.0)	94.5(3.4)
	α_M	96.5(4.6)	96.5(5.0)	96.1(4.6)	94.8(4.0)
45	α_C	96.7(4.6)	96.1(4.7)	95.4(4.6)	92.1(3.8)
	α_M	96.0(4.5)	97.3(4.8)	94.3(4.4)	91.7(3.8)

to investigate the eventual effects of FTWFC. During the study below on the detection synchronization for investigations of the FTWFC, we have also used the state α_M for comparison purposes, because, according to Ref. [7], in a maximally entangled state it is not possible to observe the effect related to the finite-time collapse. The curves in Figs. 2 and 3 are associated with these two important states for our investigation.

4. Time-scan measurements

Time resolved coincidence counts, between detectors placed at different output ports of the Sagnac, are necessary to verify the hypothesis proposed in [7]. In this way, the coincidence counts are related to the coefficient α of Eq. (5), and in an unbalanced state these coincidence counts should present a peak or a valley when the arrival time of the entangled photons in the detectors are the same, if there would be any finite time for the collapse of the wave function. However, in order to differentiate such structures a baseline needs to be established around which some residual fluctuation of the points is expected. It is not straightforward to determine such baseline due to systematic deviations coming from misalignments of the delay line as it is scanned. This has been circumvented by performing the measurement for the two states highlighted in the previous section, since both are subjected to the same misalignment but one would be affected by the FTWFC and the other not. We show then that the subtraction of the results for these two states is enough to compensate for misalignment errors, and a flat baseline is obtained as a result of this procedure.

In order to set the system for equal arrival time of the entangled photons at detectors D1T and D2T, a fraction of the laser beam at $\lambda = 800$ nm, reminiscent from the second harmonic generation process, is used (the Alignment beam in Fig. 1). This beam follows the same optical path of the down-converted photons, and its two outputs from the Sagnac are coupled into the two small, 30-cm optical fibers of Fig. 1. A Fiber Beam Splitter (FBS) is attached to the outputs of the fibers that would originally go into the two detectors, with the A (B) input of the FBS connected to the original D1T (D2T) output, as shown in Fig. 4(a). The output of the FBS is directed to a Photo-Detector (PD, type PDA36A from Thorlabs). The delay line was scanned and an interference pattern was observed at PD for a delay $\tau_{AB} = -5.47 \pm 0.03$ ps. The FBS inputs are then interchanged as shown in Fig. 4(b), and the delay line was scanned again. This time, the interference pattern was observed at $\tau_{BA} = +5.47 \pm 0.03$ ps. The two peaks, associated with the two interference patterns are shown in Fig. 4(c), where the zero is defined by $\tau_{AB} + \tau_{BA} = 0$. The difference between τ_{AB} and τ_{BA} comes solely from the difference between the FBS arms connecting the inputs A and B to its center. The zero defined through this procedure provides an estimation of the position of the delay line for which photons from the Sagnac would arrive simultaneously at D1T and D2T. The precision of this estimation is limited by variations on the distance from the fiber tip to the detector surface on D1T and D2T, respectively. We estimate that a scan from -0.4 mm (-1.3 ps) to $+0.4$ mm ($+1.3$ ps) around the zero is enough to account for any uncertainty on the location of the origin.

We expect the photons to present strong chirping in frequency due to propagation in the 30-cm optical fibers. The original 100-fs pulse duration, corresponding to the 10-nm bandwidth, should then grow to around 700 fs. This effect should limit any peak or valley observed on the scan to have a width of at least 700 fs. From this estimation, we considered steps of 130 fs of the delay line to be a good compromise between the number of points to be taken and the required precision to measure any possible structure coming from FTWFC.

After defining the step and region of the scan, coincidence counts were made as required to observe FTWFC. In Fig. 5 we present the experimental coincidence counts associated with the function:

$$F(\theta_p) \equiv \frac{C_{RT} - C_{TR}}{C_{RR} + C_{RT} + C_{TR} + C_{TT}}, \quad (8)$$

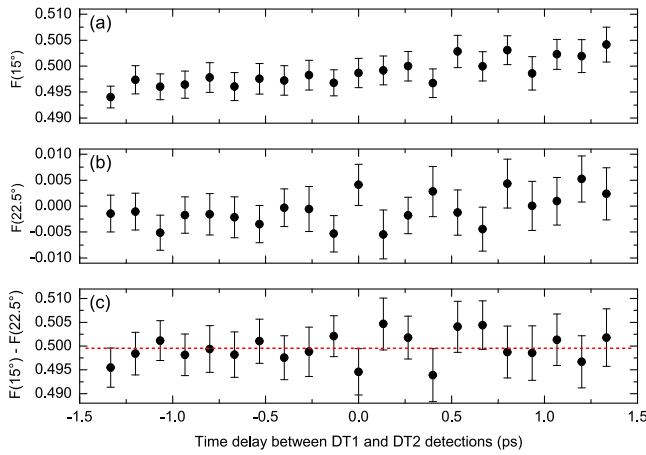


Fig. 5. (colors online) Panels (a) and (b) are averages of 48 measurements performed with a total of 300 s acquisition time per point of the $F(15^\circ)$ and $F(22.5^\circ)$ functions, respectively, as a function of the delay between the two detection channels. (c) Difference between $F(15^\circ)$ and $F(22.5^\circ)$. The red dotted line represents the mean of $F(15^\circ) - F(22.5^\circ)$. The vertical error bar is the mean standard error obtained from the 48 different realizations of the measurement.

as the delay line is scanned. For all evaluations of $F(\theta_p)$ we used $2\theta_1, 2\theta_2 = 0$, as suggested in Ref. [7]. We also set the pump QWP (ϕ_p) to zero. For these angles, we have a negligible amount of C_{RR} and C_{TT} counts, with then $(C_{RT} + C_{TR}) / (C_{RR} + C_{RT} + C_{TR} + C_{TT}) \approx 1$. The balance between C_{RT} and C_{TR} , however, will vary considerably with θ_p , as anticipated in Fig. 2. Ideally, the value of a_C proposed on Ref. [7] would lead to a near-maximum difference between these two quantities, around the value of $F(15^\circ) \approx 0.5$. On the other hand, the maximally entangled state occurring with α_M would lead to $F(22.5^\circ) \approx 0.0$.

The experimental results regarding the time scan of $F(15^\circ)$ are presented in Fig. 5(a), revealing a monotonic, small increase with the delay. Such monotonic increase, however, is also present in Fig. 5(b), which shows the corresponding experimental results when we monitor a maximally entangled state, i.e., $F(22.5^\circ)$. FTWFC is not expected to be visible in such a Bell state. In Fig. 5(c) the difference $F(15^\circ) - F(22.5^\circ)$ is presented, which, as anticipated above, is made to extract spurious effects, such as misalignment on the delay line and unbalances in fluorescence photons between different channels. The subtraction presented in Fig. 5(c), then, keeps no trace of such small increases with the delay, as can be appreciated from the scatter of points around its average, given by the dashed red line. All points differ from the average by a quantity on the order of the error bars. In Fig. 5(a) and (b) the acquisition time was 300 s for each point in a particular scan, and each curve is the average of 48 different scans taken for the same set of experimental parameters.

5. Discussion

The goal of the present experiment was to realize the synchronization procedure for observing the FTWFC effect, as theoretically proposed in Ref. [7], with the required quantum states with ultrashort wavepackets. In this way, we provide a first evaluation of the imperfections and challenges of the actual experiment, even if various aspects of the apparatus could be individually improved beyond the point employed here. Fig. 5(c) presents the main results of this first test. We were able to generate a time scan of the relative arrival time at the detectors, with steps of 130 fs, long enough to detect any peak or valley related to FTWFC and compensated for the most important systematic deviations resulting in spurious structures on the scan. In this way, it is possible to compare all points in the scan to a simple average value and demonstrate that the observed curve amounts to a random scatter of the points around this average value.

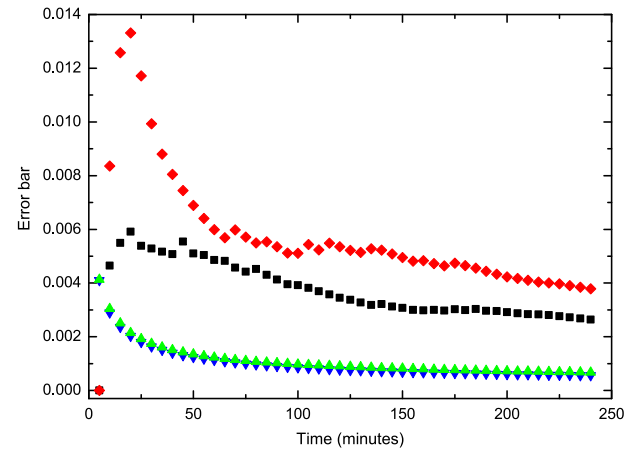


Fig. 6. (colors online) Evolution of the experimental error bars. The squares [diamonds] provide the mean standard error accumulated up to a certain measurement time for a particular point in Fig. 5(a) [5(b)], the down [up] triangles provide the respective error bars coming from the expected statistical fluctuations in the number of accumulated counts.

We can also compare the observed experimental imperfections with the theoretically predicted ones, to have an idea of how distant we still are from the point that generated the estimations of Ref. [7]. The main difference is summarized in Fig. 6 plotting the evolution of the error bars during the 48 measurements of a specific point of the curves presented in Fig. 5(a) (black circles) and 5(b) (red circles). These error bars are evaluated from variations of the measured value throughout all measurements up to a certain accumulated time. In Fig. 6 we also present the error bars associated with the expected statistical fluctuations as calculated from the square root of the total number of counts involved in the accumulated measurements. These statistical fluctuations were the main imperfection considered in Ref. [7], and the possibility to distinguish an effect related to FTWFC above this error level was an important point of the theoretical proposal. We clearly see, however, that other fluctuations on the experimental apparatus still prevail over the statistical fluctuations on the number of counts, possibly preventing the experiment to test the theoretical proposal. In principle then, according to the theoretical analysis of Ref. [7], the FTWFC effect could end up hidden between the levels of the two types of error bars in Fig. 6.

The above discussion leads to a more detailed analysis of the experiment to evaluate its imperfections and the possibility to overcome them. The first point to have in mind in this analysis is the relevant time scale of experimental fluctuations. Fig. 6 already highlights the fact that each point in Fig. 5(a) and (b) takes a total of 240 min to be evaluated. Since we have a total of 42 independent points in Fig. 5, it was necessary seven days of total measurement time, during which mechanical and thermal fluctuations in the Sagnac interferometer and delay line may affect the relative number of counts in the various detection channels. We also observed variations correlated to long term oscillations in the laser power. It is necessary then to accumulate counts over this period of time at a much higher rate, to increase the statistical error bars above the level coming from these long term fluctuations in our experimental setup. We consider that improvements along this line are completely feasible. The photon-pair generation rate in our experiment, around 160 Hz, is more than one order of magnitude smaller than the state of the art for SPDC systems [14,19,20]. There are also various improvements that can be made to better mechanically stabilize the Sagnac interferometer and the optical path of the delay lines.

These improvements will further approximate the experimental conditions to the theoretical analysis based on the FTWFC effect prevailing over the statistical errors of the system. However, there are

also improvements aiming specifically to enhance the effects related to FTWFC. Particularly, the signal related to FTWFC should increase as we narrow down the scan region where detection coincidences are expected to occur. As explained in Section 4, in our present setup we expect any peak or valley related to FTWFC to have around 700 fs, due to the chirping of the laser pulse on the short, 30-cm optical fibers on the two competing detection channels. This is considerably larger than the limit of 100 fs expected from the laser pulse temporal width. In principle, then, by removing such fibers there is room to significant improvement on the signal-to-noise ratio if FTWFC do exist. In practice, we would need also to setup some stage for chirping compensation close to the detectors in order to obtain the minimum pulse widths. This will require a major redesign of our detection system, including the scheme for single mode filtering and the protocol to find the zero delay of the time scan.

Finally, in order to test the proposal in Ref. [7] we need to synchronize all four detectors in Fig. 1. In practice, this would imply the use of three delay lines like the one for detector D_{17} . The space of scan parameters would increase by a very large amount. Assuming the present data statistics was enough to establish the base line in Fig. 5(c), we can extrapolate from there to estimate the typical numbers for the final experimental test. For example, if we would keep the same number of 42 points in the two curves in Fig. 5(a) and (b), the three delay lines would lead to the measurement of $42^3 \approx 74\,000$ points. If we were to continue spending 240 min to accumulate the necessary statistics per point, the total experiment would require 34 years! Here again it is crucial to reach the state of the art performance for SPDC systems. In this case, we estimate that the same statistics could be obtained with around 6.4 min per point, which would reduce the total acquisition time to around 11 months, still impractical. However, if we decrease the number of points in each curve by a factor of ≈ 2 , to 9 points per curve, the total amount of time would decrease by a factor of 13. This would leave us with a total time for the experiment of 25 days, which we consider feasible.

In summary, starting from the present setup, we consider the following steps crucial for the final implementation of the test for FTWFC:

1. Increase of the photon-pair generation rate to around 6000 Hz.
2. Installation of motorized delay lines in three of the four detection pathways of Fig. 1.
3. Reduction of the number of points in each time scan by a factor of 2.

Some other modifications could also be beneficial to increase the sensitivity of the final test:

1. Reduction of the size of the small detection fibers by a factor of 3, to 10 cm. This would decrease the spread in time of the photonic wavepackets, increasing the signal-to-noise ratio of a possible FTWFC effect. It would also decrease the size of the delay line, reducing the spurious pointing deviations in the time scan.

2. Free space compensation for the chirping of the detection fibers. This would improve the signal in the same direction as the reduction of the detection fibers. However, with such compensation we can in principle achieve the minimum time spreading of the photonic wavepackets, around 100 fs in our system.
3. Improvement of the visibilities in Table 1, which can reduce spurious backgrounds coming from a small amount of statistical mixture in the final polarization entangled states.

6. Conclusions

As a whole the present experimental investigation represents a large step towards testing the theoretical ideas for FTWFC introduced in Ref. [7]. We implemented a synchronization stage of two detections as required by the theoretical proposal, with the appropriate unbalanced polarization-entangled quantum states. In this way, we were able to evaluate the main imperfections of the experimental apparatus and to provide specific estimations for the amount of time required for the final test of the proposal. As a consequence, we provide a detailed description for this final experiment with the required rate of photon-pair generation to be able to succeed in the test of FTWFC. We also provide a list of improvements that could enhance the signal-to-noise ratio of the experiment, overcoming various imperfection in the present setup. This points then to the specific conditions that should be attained by any future experiment along these lines.

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